

EXPERIMENT 11: PERFORM AFFINE TRANSFORMATION ON AN IMAGE

Aim

To perform affine transformation on an image using Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy.
2. Read the input image.
3. Define source and destination points.
4. Compute affine transformation matrix.
5. Apply cv2.warpAffine().
6. Display and save the output image.

Python Code

```
import cv2
import numpy as np

# Read the input image
image = cv2.imread("input.jpg")

# Check whether image is loaded
if image is None:
    print("Error: Unable to load image.")
    exit()

# Get image dimensions
rows, cols = image.shape[:2]

# Define three source points
pts1 = np.float32([
    [50, 50],
    [200, 50],
    [50, 200]
])
```

```
# Define corresponding destination points
pts2 = np.float32([
    [10, 100],
    [200, 50],
    [100, 250]
])

# Compute Affine Transformation Matrix
matrix = cv2.getAffineTransform(pts1, pts2)

# Apply Affine Transformation
affine_image = cv2.warpAffine(image, matrix, (cols, rows))

# Save the transformed image
cv2.imwrite("affine_output.jpg", affine_image)

# Display results
cv2.imshow("Original Image", image)
cv2.imshow("Affine Transformation", affine_image)

print("Affine Transformation Matrix:")
print(matrix)

print("Affine transformed image saved as affine_output.jpg")

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the affine transformation was successfully performed on the input image using OpenCV.